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CS 5480 Final Project Report

Goal

Perform crack segmentation with sufficient accuracy and speed to allow for real time crack detection.

Impact

The ability to predict crack generation in real time offers the ability to not only save money but also improve standards of safety. Concrete construction forms the backbone of infrastructure in the modern world due to its durability, cost effectiveness, and material availability [1]. Cracks in concrete typically originate on and are observable from the surface of the structure. Being an easy to observe and reliable metric of structural health and wear, cracks are an essential metric used in structural inspections [1]. Real-time crack detection would facilitate early warning of structural failures in dams, tunnels, buildings, and roads. Automating this process would reduce human error and enable non-intrusive monitoring even in dangerous or hard-to-reach areas. The added safety made possible by these technologies has the potential to reduce loss of life and property as well as increase public trust in the safety of aging infrastructure. Real-time segmentation would enable condition-based maintenance as opposed to routine or reactive repairs. Asset owners would be able to prioritize repairs, thereby optimizing maintenance schedules and budgets. The lowered need for manual inspections would reduce the labor cost associated with the upkeep of infrastructure. Increased accuracy in structural assessment would ensure that repairs are done when and where truly needed, preventing the waste of resources on pointless over-maintenance.

Challenges

It is critical to determine both the width and number of cracks as a means of assessing structural integrity, while their orientation and spatial distribution provides insights into the causes of cracks and aids in determining whether they are structural [3]. Complex textures, varying lighting conditions, and stains or spots can make it challenging to detect cracks, and the appearance of small and narrow cracks can be indistinguishable from background noise. Finally, the quality of images can vary due to factors such as resolution, focus and capture angle, necessitating the need for robust and generalizable methods for crack inspection [2].

Literature Survey

In the effort of increasing the accuracy and speed of the crack segmentation process several developments have been made. Feng et al. (2023) used a multi-sensor fusion SLAM system enhanced by Real-ESRGAN super-resolution and Median Filter + Niblack & Connected Connectivity for crack segmentation [2]. Robustness is enhanced in noisy or low light conditions by fusing RGB cameras, IMU, and LiDAR [2]. Feng et al. allowed crack detection to be performed accurately in large scale infrastructure using mobile inspection data at significantly less time and similar performance to the existing Structure from Motion standard [2]. Deng et al. (2025) improved on this with their few-shot data segmentation pipeline aiming increasing effectiveness in practical scenarios without suffering generalization issues from specific crack datasets [1]. They utilized foundation models to improve generalization, with the base model DeepLabv3+ being trained on a big crack dataset (CrackSeg9k), and its output being refined by SAM without any fine tuning, significantly boosting segmentation recall and adaptability to new scenarios [1]. The deep learning and improved SLAM / 3D reprojection algorithms setup greatly improves automated 3D crack measurement compared to Feng's approach [1]. As a potential replacement to the old, heavyweight SAM model, Zhang et al. combine EfficientViT and SAM principles forming an optimized SAM variant for general-purpose segmentation even on mobile hardware [9]. Building on previous crack segmentation deep learning models, Liu et al. introduced SCSegamba, a lightweight structure-aware vision Mamba network for

use in crack segmentation [6]. Using a Structure-Aware Visual State Space module SCSegamba is able to capture crack morphology and texture efficiently while requiring minimal computational resources [6]. SCSegamba integrates structure-awareness to capture both linear and irregular crack patterns [6]. CrackMamba integrates the Mamba architecture with attenuation mechanisms, enhancing the model's ability to capture global contextual information while still maintaining linear spatial and computational complexity [16]. When evaluated on datasets comprising concrete pavement, asphalt pavement, and steel cracks, CrackMamba was shown to consistently improve the baseline model's performance [16]. Improvements were shown across all evaluation metrics including both IoU and mIoU, while also reducing the number of parameters and computational costs [16]. CT-CrackSeg uses BAM to improve fine-edge detection as well as boundary sharpness [17]. CT-CrackSeg architecture supports domain transfer due to its hybrid convolution-transformer structure [17]. While the lightweight design offers greater efficiency than full transformer models, it is not yet sufficiently fast to be considered fully real time [17]. CrackNex integrates Retinex theory to separate reflectance from illumination, allowing the model to focus on crack features independent of lighting variations, addressing the issue of poor lighting which is one of the largest real-world challenges in structural health monitoring [18]. CrackNex is designed to perform effectively with a limited number of annotated samples using a support-query prototype learning framework, allowing for rapid adaptation to new environment [18]. By introducing a fusion model integrating both support and reflectance prototypes CrackNex offers both increased segmentation accuracy and robustness [18]. CrackSegFormer uses Transformer-based architecture to capture multi-scale features, thereby improving the model's ability to detect cracks of various sizes and complexities [19]. CrackSegFormer enhances the model's understanding of context by focusing on semantic segmentation, allowing more accurate delineation of cracks from the surrounding surface [19]. CrackSegFormer has the added benefit of being adaptable to various surface types [19]. HrSegNet4CrackSegmentation maintains high-resolution feature maps allowing for the detection of small and thin cracks [20]. The Semantic Guidance Module merges high-level semantic information into early, detail-rich layers – increasing the model's ability to distinguish between cracks and background noise [20]. The efficient architecture of HrSegNet4CrackSegmentation allows competitive speed to be achieved without sacrificing performance [20].

Technical Gaps

Despite many promising advancements in the world of real-time crack segmentation, there are still several persistent technical gaps. While models like SCSegamba [6] and EfficientViT-SAM [9] focus on lightweight design, the gap between low-latency inference and high resolution, high accuracy segmentation has not been fully bridged. These problems are particularly pronounced in cases where lighting or environmental conditions are particularly diverse. The issue arising from this is that when using embedded systems, the high computational cost is a hinderance to real-time deployment. Many segmentation models are trained using homogeneous datasets, such as concrete surfaces in CrackFormer [10], pavements, or lab-acquired bridge images in studies [1] and [2]. This means that models are often unable to generalize across different textures, lighting situations, or material. The inability to generalize prevents, for example; a model trained using composite from performing accurately when analyzing an asphalt structure. Despite the promise shown by few-shot techniques [1] and noisy-label-aware methods [11], there are several shortcomings. Distribution-aware models are not often evaluated for real-time feasibility, leading to them being impractical for on-device use. Few-shot methods struggle when tasked with capturing the fine-grained textures of thin cracks under extreme visual variability. While sensor fusion SLAM methods, such as those put forth in studies [2], [3], and [7], are strong in spatial localization, they are not fully integrated with real-time semantic segmentation pipelines. Although Deng et al. [1] combine segmentation and precise 3D crack width/depth measurement, the 3D reconstruction pipeline is still resource-heavy and generally cannot operate on a real-time basis. This is important as quantitative information is often necessary for engineers to make structural health assessments. Feng's [2] reliance on super-resolution with Real-ESRGAN significantly reduces the speed of their pipeline. Extra SR processing can bottleneck end-to-end systems, particularly when low-power devices are used. The study conducted by Wu et al. [12] points out the lack of standardized datasets, metrics, and real-world evaluation settings, such as at night or underwater. The importance of this is that it is difficult to prove real-time capability or make fair comparisons without standard benchmarks by which performance must be assessed.

Validation and Benchmarking

To achieve real-time crack segmentation while maintaining appropriate standards of accuracy and practical deployment readiness, several comprehensive and context-sensitive metrics were used. For understanding the quality of the super-resolution model for improving the quality of input images, Peak Signal to-Noise Ratio (PSNR) measures the fidelity to the original pixel values, and Structural Similarity (SSIM) calculates the structural similarity to evaluate textural and structural information shared, such that both combined measure objective and perceptual fidelity. To evaluate the quality of the core crack segmentation model, mIoU, F1 and FPS are the primary validation metrics, with others (Precision, Recall) explaining the class imbalance. The F1 score will be used to evaluate performance while balancing precision and recall, being sensitive to class imbalance, which is common in crack datasets, and is particularly useful when cracks occupy a very small portion of the image. Precision and Recall are also important performance metrics when evaluating the efficacy of performance in real-time applications where missing a crack could have severe consequences. Throughput, measured through frames per second (FPS) measures how efficiently the model processes individual images, and is vital to assess feasibility for real time deployment, with an FPS at least above 5 FPS and ideally at least 15 FPS being important for real-time operation on edge devices. Mean IoU (mIoU) quantifies how well the model segments the target regions (e.g., cracks) [11].

To evaluate performance of pre-trained super-resolution models, the DIV2K test set was used. DIV2K is a standard dataset for super-resolution because it offers high-resolution (2K), diverse, and realistic images suitable for training and evaluation. It supports multiple degradation types (e.g., bicubic, blur, noise) and is widely used in benchmarks like the NTIRE challenges, providing a more modern and scalable testbed compared to older datasets like Set5 or Urban100. Bicubic 4x downscaling was used as the base input image. For the core crack segmentation, TUT, Crack Conglomerate and CrackSeg9k were used. TUT (from Tampere University) uses a 70/10/20 split and served as the main training dataset for custom models like SCSEgamba. Crack Conglomerate (Virginia Tech, 10,955 images merged from 9 public sources) was used for testing, but had generalization issues when trained alone. CrackSeg9k (Harvard, compiled from 10 datasets) was used by DeepLabV3+, HRSegNet, CrackMamba, SegFormer, and CrackNeXt for training. A hybrid TUT + Crack Conglomerate dataset (TUT + 80/10/10 split from Crack Conglomerate) was used to test if Crack Conglomerate would augment TUT's performance for SCSEgamba.

Method	Avg_PSNR	Avg_SSIM	Avg_Time_ms	FPS
EDSR	29.2478	0.8397	933.05	1.07
RCAN	29.2422	0.8397	548.80	1.82
NinaSR	28.5073	0.8318	137.06	7.30
CARN	28.9759	0.8327	145.19	6.89
Bicubic	26.6952	0.7663	28.95	34.55
Lanczos	26.9350	0.7740	39.61	25.25
DSCF	28.8053	0.8273	91.77	10.90
DSCF_ONNX	28.8053	0.8273	26.17	38.20
RealESRGAN_ONNX	23.8834	0.6895	449.12	2.23

Figure 1: Super resolution model comparison results

DSCLoRA was tested as the candidate SR model due to it being the winner of NTIRE 2025 competition for efficient image super resolution and SOTA model. Comparing it to standard torchsr models, torchvision's built in fsrnn and Real-ESRGAN used in Liu et al's paper, we did a comprehensive evaluation with both non compiled and onnx compiled version of the model to assess performance and quality on a V100 GPU node on the Mill on the DIV2K bicubic dataset. We found that while DSCLoRA had only the third best performance, it was only .4 PSNR lower than the top model and had 10 times faster inference. Compared to Real-ESRGAN, DSCF completely dominates with a PSNR almost 5 greater and a .13 better SSIM, in addition to running 17 times faster.

For crack segmentation at the core of the pipeline, for the sake of expediency I used 100 image subset (seed of 42) from the Crack Conglomerate Dataset test set as a benchmark, as well as using bicubic as a simple, efficient super-resolution algorithm to compare the level of model performance degradation without the hurdle of

implementing the dscv ai model into the system. I compared four different models: DeeplabV3+, trained on CCD, as the baseline, HrSegNet, trained on CrackSeg9k, as a specialized model for high resolution, YOLOv8-crack-seg, trained on the Crack Seg dataset, as a standard modern crack detection model, and SCSegamba, trained on TUT, as a new candidate model claiming to improve upon other models. For performance degradation, I compared mIoU, F1 Score, FPS, and the mIoU between original (1x), double resolution (2x), and quadruple resolution (4x), with the mIoU ratio of 2x/1x and 4x/1x being calculated for further clarity. I attempted to run CrackNex for testing, but faced memory spikes that prevented me from testing it even at 2x scale due to the model being built for small input sizes.

model	mIoU_orig	F1_orig	mIoU_2x	F1_2x	mIoU_4x	F1_4x	ratio_2x	ratio_4x	FPS_1x	FPS_2x	FPS_4x	FPS
yolov8n	0.573622	0.292545	0.55551	0.235179	0.505274	0.085022	0.968425	0.880849	23.33255	59.77743	24.57717	35.89572
yolov8s	0.713887	0.614442	0.681762	0.551853	0.566283	0.271515	0.955	0.793239	20.11352	42.58773	14.37519	25.69215
yolov8m	0.709799	0.600576	0.661855	0.50249	0.540374	0.189884	0.932453	0.761305	9.676627	22.70104	7.250352	13.20934
hrsegnet	0.555134	0.522891	0.56744	0.525808	0.524402	0.336539	1.022168	0.944641	54.71707	50.4405	40.30055	48.48604
deeplabv3+	0.694581	0.615878	0.663767	0.526983	0.583074	0.311971	0.955637	0.839461	18.49828	5.481165	1.415287	8.46491
scsegamba	0.553637	0.068991	0.503225	0.071939	0.484171	0.073981	0.908944	0.874529	20.12034	7.108418	1.845635	9.691463

Figure 2: results for crack segmentation model survey

YOLOv8s achieved the best balance of accuracy and efficiency, with an mIoU of 0.714, F1-score of 0.614, and a practical FPS of 25.7. YOLOv8m offered similar accuracy but with a noticeable drop in speed, while YOLOv8n delivered high FPS (35.9) at the cost of lower segmentation quality. DeepLabv3+ matched YOLOv8s in accuracy (mIoU 0.695, F1 0.616) but was significantly slower (FPS 8.5), reflecting its older architecture. HRSegNet provided a strong trade-off with competitive accuracy (F1 0.523) and the fastest inference (FPS 48.5), validating its high-resolution, real-time design. SCSegamba with struggled with this task (F1 0.069), indicating limited generalization without further tuning. All models showed accuracy degradation at higher scales (2x, 4x), but YOLOv8, HRSegNet and SCSegamba maintained better robustness compared to traditional CNN models.

Bonus: Model Improvements

For SCSegamba, five different attention types (GBC, ECA, GBC+ECA, SEBICA, SFA) and three different weighting schemes (original, static weighted fusion, dynamic weighted fusion), and two datasets / three hybrid datasets (TUT, TUT + Crack Conglomerate, Crack Conglomerate) were tested to improve the multi fusion strategy performance and address the lack of learnable attention for these components, making the use of SAM-based refinement unnecessary.

Dataset	Fusion	Attention	mIoU	ODS	OIS	F1	Precision	Recall	FPS
2023_Baseline	N/A	N/A	0.0144	0.0527	0.0527	0.055	0.0283	0.9997	68.24
TUT	original	gbc	0.4818	0.0562	0.0628	0.055	0.0283	0.9954	12.89
TUT	dynamic	gbc	0.484	0.0652	0.0801	0.055	0.0283	0.9951	14.32
TCC	original	gbc	0.4195	0.0616	0.0698	0.0629	0.0325	0.9963	12.95
TCC	dynamic	gbc	0.4219	0.0566	0.0646	0.0625	0.0323	0.9961	14.29
TUT	dynamic	gbc_eca	0.4832	0.0528	0.0676	0.055	0.0283	0.9957	14.26
TCC	dynamic	gbc_eca	0.4191	0.0605	0.0687	0.0629	0.0325	0.9963	14.34
TUT	dynamic	eca	0.4845	0.0543	0.0719	0.055	0.0283	0.9954	15.51
TCC	dynamic	eca	0.4197	0.0635	0.0728	0.063	0.0325	0.9959	15.49
TUT	dynamic	sebica	0.4841	0.0579	0.0714	0.055	0.0283	0.9957	14.75
TCC	dynamic	sebica	0.4194	0.0579	0.0636	0.0629	0.0325	0.9944	14.57
TUT	dynamic	sfa	0.4852	0.0672	0.0849	0.055	0.0283	0.9946	14.66
TUT	dynamic	sfa	0.4202	0.0575	0.0615	0.0628	0.0324	0.9959	14.57
Crack_Conglomerate	original	gbc	0.4183	0.0583	0.0638	0.063	0.0325	0.996	12.74
Crack_Conglomerate	weighted	eca	0.43	0.0539	0.0571	0.0614	0.0317	0.9953	15.09
Crack_Conglomerate	weighted	gbc_eca	0.4291	0.0554	0.0572	0.0616	0.0318	0.9956	14.02
Crack_Conglomerate	weighted	sebica	0.4258	0.0573	0.0608	0.062	0.032	0.9958	14.99
Crack_Conglomerate	weighted	sfa	0.4339	0.0555	0.0635	0.0609	0.0314	0.9957	14.95

Figure 3: Results for experiment to improve SCSegamba single-shot performance and real-time capability.

The dataset trained on the TUT dataset with the SFA attention has the highest mean IoU score, but ECA is only .007 lower and is faster by 1 frame per second in evaluation. ECA 2.62 frames per second faster and has .0027 higher mean IoU, and is moderately faster and slightly higher performance The Crack Conglomerate dataset, despite having eight times as many images and also being the source of evaluation has worse results than the TUT dataset, suggesting that the images in the Crack Conglomerate dataset may be of worse quality than the TUT dataset. While the model has very high recall, the low precision suggests that it is labelling too many pixels as cracks despite that not being the case, potentially suggesting issues with softmax / sigmoid thresholds are something along those lines.

Note of External Aid

Previous drafts of this report, done as part of the CS6604 class were a 2 person group project done in conjunction with Reece Watson. I did all the code / technical work and did the vast majority of drafting the report, but Reece contributed to minor editing and drafting his section (3D reprojection) that was not used in this report. ChatGPT was used in helping to draft the project phase 1, phase 2, and final report for the class, which served as the basis for this report, and was also used to develop the code used within the project. Feng et al., as well as Deng et al.'s papers were major inspirations for the basis of the pipeline architecture used in the CS6604 report, of which this project focuses only on improving the super-resolution and single shot crack-segmentation aspects and not SLAM, improvement with SAM or reprojection in 3D, which is beyond the scope of this project.

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